

HelixVPN — Master Decision Register (expanded)

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Revision: 3 **Last modified:** 2026-07-04T12:00:00Z **Status:** active — Volume 0 (Spine, meta & governance) nano-detail document

Rev 3 (2026-07-04 hardening pass): Independently re-verified this whole register against SPECIFICATION.md, 99-source-coverage-ledger.md, and every owning volume doc during a full-corpus hardening pass — no contradictions found (this register was already internally consistent and accurate; a genuinely rare result across the corpus). One clarifying cross-reference added: D-PKI-CA-TIER's owning doc (v05-security/pki-and-certs.md §2.1) now composes with a **new operational distinction** added to v03-control-plane/svc-pki.md during this pass — CA rotation is split into a *scheduled* path (24h trust-overlap, routine) and a *compromise* path (immediate, zero-overlap, force-revoke-and-re-enroll) — which sharpens (but does not change) this register's D-PKI-CA-TIER reversal criterion ("operational evidence the single online issuing CA is a bottleneck"): a *compromise* event is now handled as its own incident path distinct from a *bottleneck* reversal trigger, so the two are no longer conflated in the implementation the way they briefly were before this pass. No decision status changed. **Rev 2:** Added the §0-mandated explicit reversal line to the seven §4 decisions that lacked one (D-AC-1, D-OPENAPI-AUTHORING, D-CODEGEN-TRACK, D-K8S-EDGE-INGRESS, D-K8S-PG-HA, D-GW-SELECT, D-DESIGN-EXTRACT); fixed the §1 D4 "Resolved by" cross-ref doc 04/svc-ipam.md → v03-control-plane/svc-ipam.md (svc-ipam.md is a Volume-3 doc).

Authority: Subordinate to ../SPECIFICATION.md §9 (the spine decision register D1-D8, which this document expands). Where this register disagrees with the spine on a spine decision's camps or recommendation, the spine wins until amended per §11.4.73. The expansion-surfaced decisions (the D-... family) are owned by their volume docs and consolidated here per §11.4.66/.101 (no decision silently resolved).

Document role. The single consolidated decision register for the whole HelixVPN final/ set. It (1) expands the eight spine decisions **D1-D8** with the full options, rationale, the gate/evidence that resolves each, and reversal criteria; and (2) catalogues every **expansion-surfaced decision** (D-...) the nano-detail volumes raised — PKI CA tiering, audit enum, codegen tracking, K8s ingress/PG-HA, gateway selection, the OpenDesign reconciliation, the repo/design extraction timing, and the per-platform/per-client/per-FFI choices — each with its owning doc, recommendation, and reversal trigger. Per §11.4.6 no decision is presented as settled when the spine marks it open; per §11.4.101 an irreversible / high-blast-radius decision is operator-gated and named as such.

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0. How a decision is recorded

Per §11.4.66 (interactive-clarification / options-not-free-text) and §11.4.101 (autonomous-decision-over-blocking), every open or surfaced decision carries six fields:

Field	Meaning
ID	Dn (spine) or D-... (expansion-surfaced), stable.
Decision	the one-sentence question.
Options	the mutually-exclusive camps/choices (Camp A / Camp B / a/b/c).
Recommendation	the current best call (which may be "defer", "both", or a safe default per §11.4.101).
Resolved by	the named gate (G1-G6), benchmark, or owning doc + operator trigger that converts the recommendation into a binding fact.
Reversal criteria	

Field	Meaning
	the captured evidence that re-opens a settled call (§11.4.6 — a settled call is never silently un-settled; §11.4.7 demotion needs same-conditions evidence).

A recommendation is **not** a resolution: a decision is resolved only when its *Resolved by* gate produces captured evidence (§11.4.5/.69/.107) or its operator trigger fires. Decisions with an irreversible / high-blast-radius action (D8 licensing, repo creation, force-history) are operator-gated (§11.4.101 / §11.4.113).

1. Spine decisions D1-D8 (the master register)

Reproduced from [SPEC §9] with the resolution gate and reversal criteria made explicit. The camps are the source-research camps ([99] decision cross-check).

#	Decision	Camp A	Camp B	Recommendation
D1	Primary obfuscating transport	MASQUE/QUIC = WG-over-HTTP-3 (RFC 9298/9297/9221) — Mullvad's actual mechanism	Hysteria2 + Salamander turnkey QUIC+obfs primary, WG fallback	MASQUE primary for true Mullvad parity + single Rust impl; keep Hysteria2 knobs as a quinn reference
D2	Shared client-core language	Rust core + Flutter UI	Go core + Flutter UI (reuses Hysteria2 Go)	Rust — wins on iOS NE memory ceiling + WASM
D3	Event bus	Redis Streams (MVP) → NATS JetStream (scale)	NATS JetStream from start	Redis Streams MVP, NATS Phase 2 — the taxonomy is bus-agnostic, swap is transport-only

#	Decision	Camp A	Camp B	Recommendation
D4	Overlapping-RFC1918 collision across N nets	IPv6 ULA /48 per tenant + 4via6	CGNAT 100.64/10 1:1 per network	4via6 for scale + clean overlay; hide the mapping behind Console UX (a v1 must-decide)
D5	Gateway-edge language (MASQUE termination)	Rust (quinn+h3, shares helix-transport byte-for-byte)	Go (quic-go + masque-go turnkey)	Rust if G4 holds — single-implementation guarantee; else Go acceptable
D6	Transport topology	single protocol end-to-end	asymmetric per-leg (Hysteria2/QUIC user↔gateway, WG gateway↔networks)	surface MST's best-fit-per-leg as a Phase-2 option ; MVP keeps WG end-to-end with pluggable transport
D7	MVP ambition	lean tunnel-first (CLI 2-host → API → apps)	full ecosystem / "Connectivity-OS" from v1	Lean Phase-0 spike then MVP ; defer the ecosystem to Phase 2/3

#	Decision	Camp A	Camp B	Recommendation
D8	Licensing / positioning	source-available + commercial	pure OSS	decide before public release; same code serves self-host + managed

2. Spine decisions — per-decision detail

D1 — Primary obfuscating transport

- **The question.** Which transport is the *primary* censorship-evasion carrier under WireGuard?
- **Why it matters.** This is the single decision that defines whether HelixVPN reaches genuine Mullvad parity. The pivotal correction ([04_ARCH §1]): **Mullvad's "QUIC mode" is not a separate protocol — it IS WireGuard-over-MASQUE/HTTP-3.** Choosing MASQUE is choosing Mullvad's *actual* mechanism, not an analogue.
- **Camp A (MASQUE/QUIC, recommended).** WG datagrams ride RFC 9298 CONNECT-UDP over HTTP/3, framed to look like web traffic. One Rust helix-transport impl serves client + edge. (v02-data-plane/transport-masque-quic.md)
- **Camp B (Hysteria2+Salamander).** A turnkey QUIC+obfs protocol the plurality of the 10 analyses favoured for speed-to-ship; HelixVPN keeps its tuning knobs as a quinn reference rather than adopting the protocol. (08; 99)
- **Resolved by.** Gate **G2** — MASQUE survives an nftables UDP block at $\geq 50\%$ of plain-WG throughput (UNVERIFIED until captured). (06; HVPN-NFR-002)
- **Reversal.** A captured G2 failure to clear $\geq 50\%$ *and* a head-to-head where a Hysteria2-tuned quinn build clears the bar the MASQUE framing could not, under the same DPI sim (§11.4.7 same-conditions evidence).

D2 — Shared client-core language

- **The question.** Rust or Go for the helix-core shared by client + connector + edge?
- **Why it matters.** The iOS NEPacketTunnelProvider memory ceiling (historically ~15 MB) is the hardest constraint in the whole product (HVPN-NFR-500); Go's runtime/GC footprint is the risk. WASM reach (browser MASQUE proxy) also favours Rust.
- **Camp A (Rust, recommended; CLD chose it).** Fits the NE ceiling, compiles to WASM, gives byte-for-byte client↔edge reuse. (03)
- **Camp B (Go).** Reuses Hysteria2's Go implementation, single-language with the control plane (DSK/GMI/GCT). (99)
- **Resolved by.** Gate **G3** (make-or-break) — on-device RSS profile shows $\geq 30\%$ headroom under the NE ceiling. The overview names G3 as the literal decider for D2.
- **Reversal.** G3 captured failure: if Rust cannot fit the NE ceiling with headroom, **D2 and the entire client strategy re-open** (the program's pivot, HVPN-NFR-500).

D3 — Event bus

- **The question.** Redis Streams or NATS JetStream for the real-time event backbone?
- **Why it matters.** Convergence ($p99 < 1s$) rides the bus; but the bus is behind a transport-agnostic interface, so the choice is a transport swap, not a redesign.
- **Camp A (Redis MVP → NATS scale, recommended).** Redis Streams + consumer groups + XAUTOCLAIM DLQ for the single-node MVP; swap to NATS JetStream at multi-region scale. (v03-control-plane/svc-events.md; restated at the wiring layer in architecture-and-wiring.md §)
- **Camp B (NATS from start).** DSK/KMI; avoids the later swap at the cost of MVP heft.
- **Resolved by.** doc 02 fixes the MVP bus; a Phase-2 bus-swap integration test (HVPN-NFR-108) proves the interface is genuinely transport-agnostic.
- **Reversal.** A homelab-scale Redis Streams build that cannot hold the convergence SLO would pull NATS earlier (captured load evidence).

D4 — Overlapping-RFC1918 collision

- **The question.** How does one user reach N networks that all expose e.g. 192.168.1.0/24 without collision?
- **Camp A (ULA /48 + 4via6, recommended).** Each tenant gets an IPv6 ULA /48; advertised IPv4 LANs map into it via 4via6, so overlapping ranges become distinct address spaces; per-network NAT is the documented fallback (FR-303)

provisions it). (v02-data-plane/routing-and-addressing.md; v03-control-plane/svc-ipam.md)

- **Camp B (CGNAT 100.64/10 1:1).** GMI/KMI; a per-network 1:1 map in the shared CGNAT space.
- **Resolved by.** doc 01 (routing) + svc-ipam.md + an MVP multi-connector overlap test (FR-302). A **v1 must-decide** — the recommendation is binding for MVP.
- **Reversal.** 4via6 proves inadequate on a target platform/OS → fall back to the already-provisioned per-network NAT path.

D5 — Gateway-edge language

- **The question.** Rust or Go to *terminate* MASQUE at the gateway edge?
- **Camp A (Rust, recommended if G4 holds).** quinn+h3 shares the client's helix-transport byte-for-byte (one obfuscate/de-obfuscate implementation).
- **Camp B (Go).** quic-go+masque-go turnkey, faster to integrate.
- **Resolved by.** Gate **G4** — Go-vs-Rust edge head-to-head (throughput/CPU/p99). Rust wins unless Go is materially ahead.
- **Reversal.** G4 shows Go materially ahead *and* the single-impl reuse benefit is judged not worth the gap (a captured-benchmark + judgment call).

D6 — Transport topology

- **The question.** One protocol end-to-end, or best-fit per leg?
- **Camp A (single protocol, MVP).** WG end-to-end with the pluggable transport.
- **Camp B (asymmetric per-leg, Phase-2 option).** Hysteria2/QUIC user↔gateway, plain WG gateway↔connector (11_MST's largest analysis).
- **Resolved by.** doc 01 + svc-coordinator.md per-leg transport policy (the coordinator already sets per-node transport policy, so per-leg is a Phase-2 config, not a redesign).
- **Reversal.** A per-leg deployment measurably beats end-to-end WG under real DPI without breaking the single-helix-transport reuse guarantee.

D7 — MVP ambition

- **The question.** Lean tunnel-first MVP, or full "Connectivity-OS" from v1?
- **Recommendation.** Lean Phase-0 spike → MVP; defer the ecosystem to Phase 2/3 (the roadmap §8 is the resolution).
- **Reversal.** Operator re-scopes the program (a §11.4.66 scope decision, not an evidence trigger).

D8 — Licensing / positioning

- **The question.** Source-available + commercial, or pure OSS?
 - **Recommendation.** Decide before public release; the same code serves self-host + managed either way ([04_ARCH §13]).
 - **Resolved by.** A **pre-Phase-3 operator decision** — irreversible/high-blast-radius once public, so operator-gated per §11.4.101 (the agent does not pick a license autonomously).
 - **Reversal.** Effectively one-way once the public license ships; hence the pre-release gate.
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3. Expansion-surfaced decisions (the D-... family) — index

These were raised by the nano-detail volume docs during expansion and are owned by those docs; they are consolidated here (never silently resolved) per §11.4.66/.101. Each composes with — and none overrides — the spine register D1-D8.

ID	Decision	Recommendation	Owning doc
D-PKI-CA-TIER	CA topology	single online issuing CA under an offline/KMS root (spine option B)	v05-security/pki-and-certs.md §2.1
D-AC-1	auth.login/ auth.enroll.denied audit rows	reconcile-before-tag: pick add-to-enum vs defer, align master §7	v05-security/audit-and-compliance.md §13.1
D-AC-2	tamper-evident audit hash chain	Phase-2 additive (append-only grant is the MVP floor)	v05-security/audit-and-compliance.md §13.1
D-AC-3	audit retention default	no auto-prune (operator-driven)	v05-security/audit-and-compliance.md §13.1
D-AC-4	SIEM connector	generic REST/WS in MVP; named connectors Phase-2	v05-security/audit-and-compliance.md §13.1
D-KS-1	off-tunnel DoH-on-443 posture	accept default + document the split-tunnel residual; MDM disable for managed	v05-security/kill-switch-and-dns-leak.md §12.1
D-KS-2			

ID	Decision	Recommendation	Owning doc
	Android lockdown enforcement	both layers (core kill-switch always on + surface OS-lockdown prompt)	v05-security/kill-switch-and-dns-leak.md §12.1
D-KS-3	iOS extension-restart leak guarantee	measure on-device, mark UNVERIFIED until proven (G3/SC)	v05-security/kill-switch-and-dns-leak.md §12.1
D-CODEGEN-TRACK	track vs gitignore generated clients	gitignore; make gen is the §11.4.77 regeneration mechanism	v06-deploy/codegen-pipeline.md
D-OPENAPI-AUTHORING	spec-first vs code-first OpenAPI	spec-first (hand-authored contract; handlers validate against it)	v06-deploy/codegen-pipeline.md
D-CODEGEN-NET	codegen offline-from-clean-clone	vendor pinned local protoc-gen-*; buf generate offline after make gen-tools	v06-deploy/codegen-pipeline.md
D-K8S-EDGE-INGRESS	how UDP/443 reaches the edge on K8s	DaemonSet + hostNetwork:true (node IP = gateway IP), NET_ADMIN-only	v06-deploy/kubernetes.md §; [05 §7.3]
D-K8S-PG-HA	Postgres HA on K8s	Patroni/ CloudNativePG (a) for self-managed; managed PG (b) otherwise; single StatefulSet (c) = dev-only	v06-deploy/kubernetes.md §; ha-and-multiregion.md §3
D-GW-SELECT	multi-region gateway selection	geoDNS (a) for Phase-2 MVP; anycast (b) for latency-critical; client-side from map (c) as complement	v06-deploy/ha-and-multiregion.md §; [05]
D-LLM-POLICY	ship LLM-assisted policy authoring?	defer to Phase 2; default no until product-validated	v06-deploy/repo-layout-and-decoupling.md §; helix-ecosystem-integration.md
D-REPO-CREATE		one batch at Phase-1 exit	

ID	Decision	Recommendation	Owning doc
	when to extract reusable vasic-digital repos	(shared prefix, stable seams)	v06-deploy/repo-layout-and-decoupling.md §10
D-DESIGN-EXTRACT	when helix_design decouples	decoupled day-one (Volume-10 mandate, independent of code churn)	v06-deploy/repo-layout-and-decoupling.md §10
D-OD-1	OpenDesign-as-dependency vs authoring layer	OpenDesign = mandatory design <i>authoring/refinement</i> layer; helix_design owns the canonical tiered JSON tokens + polyglot exporters	v10-design/opendesign-foundation.md; REFINEMENT_NOTES
D-CORE-, D-FFI-, D-CLIENT-, D-UI-	Rust-core / FFI / client / UI sub-choices	per-doc (Volume 4)	v04-client/helix-core-rust.md, ffi-surface.md, helix-ui-flutter.md
D-APPLE/ ANDROID/ WIN/LXN/ OHOS/AUR-*	per-platform shim choices	per-doc (Volume 4 shims)	v04-client/shim-*.md
D-DAITA-A	DAITA via maybenot	adopt maybenot state machine above WG	v02-data-plane/daita.md
D-WG-PATH / D-SUBSTRATE-DEFAULT	kernel vs userspace WG per platform; substrate default	per-platform (kernel where available, boringtun fallback)	v02-data-plane/wireguard-core.md; architecture-and-wiring.md

Honest scope note (§11.4.6). The grep across the volumes surfaces additional doc-local D-... tags (e.g. D-WEB-*, D-ID*, D-S-1, D-T-1) that are *intra-doc* design callouts rather than program-level decisions; they remain owned by their docs and are not promoted here unless they cross a doc boundary or gate a phase. The table above is the consolidated cross-cutting set; the per-doc callouts stay authoritative in their owning doc (§11.4.35).

4. Expansion-surfaced decisions — per-decision detail

D-PKI-CA-TIER — CA topology (two-tier issuing CA)

- **Question.** Flat single-CA, or a two-tier (offline/KMS root → online issuing intermediate)?
- **Recommendation (MVP default).** A **single online issuing CA under an offline/KMS root** — i.e. spine option B (two-tier, root offline, short-lived online intermediate). The `ca_chain` delivered at enroll has length ≥ 1 so clients pin a chain from day one; promoting topology (adding a second regional intermediate) is *additive* — the chain grows, never reshapes (§11.4.6). (v05-security/pki-and-certs.md §2.1; carried from 04-security-privacy-pki.md §4.2)
- **OPERATOR CONFIRMED (2026-06-26).** Two-tier issuing CA accepted as the MVP default. Proceed with the recommendation.
- **Resolved by.** v05-security/pki-and-certs.md is binding; an integration test proves cross-tenant cert validation fails (FR-108) and chain pinning holds.
- **Reversal.** Operational evidence the single online issuing CA is a throughput/availability bottleneck → add a second intermediate (additive, no client break).

D-AC-1..4 — audit enum & audit surface

- **D-AC-1 (open, reconcile-before-tag).** Whether `auth.login / auth.enroll.denied` are rows in the *closed* audit-action enum (MVP) or deferred to Phase 2. The closed enum (svc-telemetry §4.2) is the mechanical truth; one option must be picked and master §7 aligned to it **before the release tag** (a §11.4.120 fix-breaks-its-own-gate-class reconciliation). Owning doc: v05-security/audit-and-compliance.md §13.1. **Reversal:** n/a — a reconcile-before-tag process choice; once the enum↔master-§7 alignment is picked and tagged, re-opening requires a new audit-action requirement (a §11.4.66 decision).
- **D-AC-2 (recommended: Phase-2).** Tamper-evident audit **hash chain** — the `meta.prev_hash` seam is reserved in MVP but the capability is *not claimed* for MVP (append-only REVOKE UPDATE, DELETE is the MVP floor). Reversal: a compliance requirement pulls it into MVP.
- **D-AC-3 (recommended: A).** Audit **retention** default = no auto-prune (operator-driven); control-action audit is low-volume + accountability-valuable, so the operator chooses a period per their regime. Reversal: storage pressure or a regime mandating a default TTL.
- **D-AC-4 (recommended: B).** **SIEM** connector — MVP exposes generic REST/WS surfaces; a named connector is

Phase-2 additive. Reversal: an operator standardising on a specific SIEM pre-Phase-2.

D-KS-1..3 — kill-switch / DNS-leak posture

- **D-KS-1 (recommended: A default, B managed).** Off-tunnel **DoH-on-443**: accept it inside a full tunnel (not an underlay leak) and *document the split-tunnel residual* by default; offer an enterprise DoH-disable control on MDM platforms. The residual is documented, never claimed closed (§6.4 / §11.4.6).
- **D-KS-2 (recommended: both layers).** Android **lockdown**: core kill-switch always on **plus** surfacing the OS Always-on-VPN/lockdown prompt; never silently assume the OS backstop is present.
- **D-KS-3 (recommended: B).** iOS **extension-restart leak window**: do not claim leak-proof; measure on-device (gate G3/SC) and mark UNVERIFIED until the pcap exists. Owning doc: v05-security/kill-switch-and-dns-leak.md §12.1.

D-CODEGEN-TRACK / D-OPENAPI-AUTHORING / D-CODEGEN-NET — codegen

- **D-OPENAPI-AUTHORING (binding).** OpenAPI is **spec-first** (hand-authored contract; handlers validated against it), not code-first from handler annotations — so the TS/Dart app clients generate from a stable surface. (v06-deploy/codegen-pipeline.md) **Reversal**: n/a — an authoring-workflow process choice; re-opens only if spec-first authoring captures sustained friction on a surface, reverting that surface to code-first.
- **D-CODEGEN-TRACK (binding).** Generated clients are **gitignored**; make gen from a clean clone reproduces them deterministically (the schema is the single source of truth; a committed generated tree is a second copy that rots). A .gitignore-meta/<slug>.yaml declares the §11.4.77 regeneration mechanism. This is a *visible reconciliation* (§11.4.35) of the overview's "generated code gitignored" intent into the binding Volume-6 decision. **Reversal**: n/a — a process choice; schema-as-SSoT + the §11.4.77 regen make tracking generated trees strictly worse, so no evidence re-opens it.
- **D-CODEGEN-NET (recommended).** buf remote plugins need network; per §11.4.77 the regen must work from a clean clone, so vendor pinned local protoc-gen-* plugins under tools/ (make gen runs offline after a one-time make gen-tools). **Reversal**: none expected; it strengthens the §11.4.77 guarantee.

D-K8S-EDGE-INGRESS / D-K8S-PG-HA — Kubernetes

- **D-K8S-EDGE-INGRESS (recommended: a).** A normal pod behind ClusterIP/L7 ingress **cannot** carry UDP/443 (most cloud L7 ingress can't forward UDP; an extra NAT hop breaks the WG endpoint model). The edge therefore runs as a **DaemonSet with hostNetwork:true** so UDP/443 lands directly on the node IP (node IP = gateway IP), NET_ADMIN-only, everything else dropped — parity with the quadlet model. Alternatives (record, not silently dropped): NodePort/LoadBalancer-UDP/ Gateway-API-UDP. (v06-deploy/kubernetes.md; [05 §7.3])
Reversal: a target K8s platform forbids hostNetwork/NET_ADMIN pods (captured platform constraint) → fall back to the recorded NodePort/LoadBalancer-UDP / Gateway-API-UDP ingress path.
- **D-K8S-PG-HA (recommended: a/b).** (a) Patroni operator / CloudNativePG (primary + ≥ 2 replicas, auto-failover) for self-managed fleets — keeps failover in-cluster; (b) external managed Postgres referenced by DATABASE_URL; (c) single-replica StatefulSet is explicitly **not** production-grade (the single-node parity build only). (v06-deploy/kubernetes.md; ha-and-multiregion.md §3) **Reversal:** an operator standardises on managed Postgres (b), or a Patroni/CNPG drill captures inadequate in-cluster failover → switch tiers (the (a)/(b)/(c) set is the menu; the pick is operator/evidence-driven, §11.4.66).

D-GW-SELECT — multi-region gateway selection

- **Options.** (a) **geoDNS** — name resolves to nearest region's edge IP by client geo; health-checked DNS removes a dead region; simplest, DNS-TTL-bound failover. (b) **Anycast** — one advertised IP announced (BGP) from every region; fastest failover, needs BGP/anycast capability. (c) **Client-side from the network map** — coordinator hands each node a region-specific GatewayInfo.endpoint.
- **Recommendation.** (a) geoDNS for the Phase-2 MVP (no BGP dependency); (b) anycast for latency-critical fleets with the capability; (c) as a complement — the coordinator already knows each node's region, so it can bias GatewayInfo regardless. On a regional failover the coordinator emits gateway.failover {from,to} and pushes a GatewayInfo-only delta. (v06-deploy/ha-and-multiregion.md)
- **Reversal.** geoDNS (a) DNS-TTL failover captures too-slow recovery for a latency-critical fleet → adopt anycast (b) where the BGP/anycast capability exists; the (a)/(b)/(c) options are complementary, so the switch is additive (no client break).

D-DESIGN-EXTRACT / D-REPO-CREATE — extraction timing

- **D-DESIGN-EXTRACT (recommended: b).** `helix_design` decouples **day-one** as its own vasic-digital submodule (the Volume-10 §11.4.162 mandate), independent of code churn — unlike the six code pillars (D-REPO-CREATE). **Reversal:** n/a — a process/timing choice fixed by the Volume-10 §11.4.162 day-one-decoupling mandate; re-opening requires a §11.4.26 constitution amendment.
- **D-REPO-CREATE (recommended: b; operator-gated creation).** Extract the reusable code repos in **one batch at Phase-1 exit** (shared release prefix, stable seams). Remote repo creation (`gh repo create / glab`) is irreversible/high-blast-radius and therefore **operator-gated** per §11.4.101 (the agent does not create org repos autonomously). **Reversal:** none — the timing is a process choice.

D-OD-1 — OpenDesign reconciliation (operator decision, safe default taken)

- **The finding.** Live web research (§11.4.99) found `nexu-io/open-design` is a **local-first generative-design authoring/refinement app** (DESIGN.md + tokens.css + od CLI + MCP) — **not** an npm token-dependency, and it does **not** emit Dart/Swift/Compose/ArkTS/C-Qt. This does not match §11.4.162's literal "install as a project dependency / use its design tokens" phrasing.
 - **Safe default adopted (reversible — docs only, §11.4.101).** OpenDesign = the mandatory design *authoring/refinement* layer; the decoupled vasic-digital/`helix_design` submodule owns the canonical tiered JSON token source + the polyglot exporters OpenDesign cannot produce. §11.4.162's **intent** (OpenDesign as the design system) is satisfied.
 - **OPERATOR CONFIRMED (2026-06-26).** Option (a) confirmed — the safe default interpretation is accepted. OpenDesign as the authoring/refinement layer + `helix_design` as the token source + exporter is the approved architecture.
 - **Owning doc / record.** `v10-design/opendesign-foundation.md`; `../REFINEMENT_NOTES.md` D-OD-1.
 - **Reversal.** Operator picks (b) or (c); the docs-only default makes reversal cheap.
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5. The gate → decision resolution map

The Phase-0/1 gates are the evidence engine that converts a *recommendation* into a *resolution*. This is the reverse index of §1's "Resolved by" column.

[MERMAID RENDER FAILED — block #1]

Gate / trigger	Decision it resolves	Evidence kind
G2	D1 (MASQUE primary)	throughput-under-DPI capture
G3	D2 (Rust core) — make-or-break	on-device RSS profile
G4	D5 (edge language)	head-to-head throughput/CPU/p99
G5	D-FFI-*	Flutter FFI status-stream log
MVP multi-connector overlap test	D4 (4via6)	distinct-address-space reach capture
Phase-2 bus-swap integration test	D3 (Redis→NATS)	bus-agnostic interface proof
v05-security/pki-and-certs.md integration	D-PKI-CA-TIER	cross-tenant validation FAIL + chain-pin
pre-Phase-3 operator decision	D8 (licensing), D-REPO-CREATE	operator action (not autonomous)
reconcile-before-tag	D-AC-1	enum↔master-§7 alignment

6. Reversal-criteria summary (when a settled call re-opens)

Per §11.4.6 (no silent un-settling) + §11.4.7 (demotion needs same-conditions evidence) + §11.4.114 (last-known-good regression isolation), a resolved decision re-opens **only** on captured evidence under the conditions that resolved it:

- **D1 → re-open** if G2 captured-fails $\geq 50\%$ *and* a Hysteria2-quinn build clears the same DPI sim.
- **D2 → re-open** if G3 captured-fails the iOS NE headroom (re-opens the whole client strategy).
- **D3 → pull NATS earlier** if Redis Streams captured-fails the convergence SLO at MVP scale.

- **D4 → NAT fallback** if 4via6 captured-fails on a target platform (FR-303 fallback already provisioned).
- **D5 → Go edge** if G4 captures Go materially ahead and reuse is judged not worth the gap.
- **D6 → per-leg** if a per-leg build captured-beats end-to-end WG under real DPI without breaking single-helix-transport reuse.
- **D-PKI-CA-TIER → add intermediate** if the single issuing CA captured-bottlenecks (additive, no client break).
- *D-AC-2/3/4, D-KS- → pull into MVP** on a compliance/operator requirement (a §11.4.66 decision, not autonomous).
- **D8, D-REPO-CREATE, D-OD-1 → operator** — these are operator-gated; the agent never flips them autonomously (§11.4.101).

§11.4.6 closing note. No decision in this register is presented as resolved ahead of its gate: D1/D2/D5/D-FFI await G2/G3/G4/G5 (all UNVERIFIED until captured); D4 is a v1-must-decide whose recommendation is binding for MVP with a provisioned fallback; D8 + the repo/OpenDesign calls are operator-gated. A recommendation is a *plan*, not a *finding*.

Sources verified

- *../SPECIFICATION.md* §9 (decision register D1-D8: camps, recommendations, "resolved by"), §8 (the phase gates G1-G6 that resolve D1/D2/D5).
- *../99-source-coverage-ledger.md* §"Decision-coverage cross-check" (D1-D8 camp sources) + gaps G1/G2/G3 (rejected sing-box/Fyne, DR/RTO).
- *../REFINEMENT_NOTES.md* F2 (proto unification), F3 (D8 added), R5 (subtask tiers), D-OD-1 (OpenDesign reconciliation + the three operator options).
- *v05-security/pki-and-certs.md* §2.1 (D-PKI-CA-TIER: single online issuing CA under offline/KMS root, $ca_chain \geq 1$, additive promotion); *audit-and-compliance.md* §13.1 (D-AC-1..4); *kill-switch-and-dns-leak.md* §12.1 (D-KS-1..3).
- *v06-deploy/codegen-pipeline.md* (D-OPENAPI-AUTHORING, D-CODEGEN-TRACK, D-CODEGEN-NET); *kubernetes.md* (D-K8S-EDGE-INGRESS DaemonSet+hostNetwork, D-K8S-PG-HA Patroni/CNPG); *ha-and-multiregion.md* (D-GW-SELECT geoDNS/anycast/client-side); *repo-layout-and-decoupling.md* §10 (D-REPO-CREATE, D-DESIGN-EXTRACT, D-LLM-POLICY).
- *v10-design/opendesign-foundation.md* (D-OD-1); the volume D-tag grep across v02/v03/v04/v05/v06/v10 for the surfaced-decision index (§3).

Constitution bindings: §11.4.44 (revision header), §11.4.6 (no decision resolved ahead of its gate; recommendations ≠ findings), §11.4.66 (options-not-free-text), §11.4.101 (irreversible/high-blast-radius decisions operator-gated — D8/D-REPO-CREATE/D-OD-1),

§11.4.7/.114 (reversal needs same-conditions / known-good evidence), §11.4.73 (spine wins on conflict until amended), §11.4.35 (per-doc callouts stay authoritative in their owning doc), §11.4.65/.153 (HTML+PDF[+DOCX] exports follow in refinement).